

BLOOD: Comprehensive Study Guide

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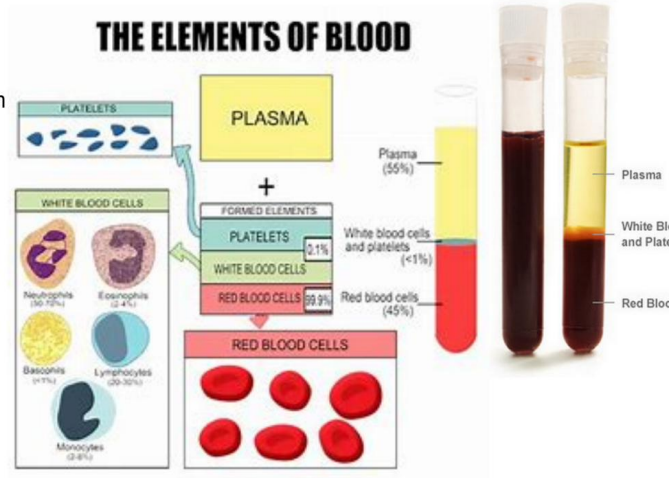
§1 · Blood Overview

Physical Characteristics

- **Color:** Scarlet (O₂-rich / oxygenated) · Dark red (deoxygenated)
- **Texture:** Sticky, opaque, metallic/salty taste; ~5× more viscous than water
- **pH:** 7.35 – 7.45 (slightly alkaline)
- **Volume:** Males 5–6 L · Females 4–5 L
- **Classification:** Fluid connective tissue — living formed elements in non-living plasma matrix

Blood

- Blood is a connective tissue composed of a liquid medium called **plasma** in which solid components are suspended.
- The **Cellular Components** of blood include:
 1. Red blood cells (RBC, **erythrocytes**)
 2. White blood cells (WBC, **leukocytes**)
 3. Platelets (**thrombocytes**)



▲ Blood composition — plasma + formed elements

8 Functions of Blood

Category	Function	Key Detail
Distribution	O ₂ delivery	RBCs carry O ₂ from lungs → all body cells
Distribution	Waste removal	CO ₂ → lungs; BUN/Creatinine → kidneys
Distribution	Hormone transport	Endocrine glands → target organs via blood
Regulation	Temperature	Superficial vessel dilation radiates heat; constriction retains heat
Regulation	pH balance	Plasma proteins = buffers; HCO ₃ ⁻ = alkaline reserve (7.35–7.45)
Regulation	Fluid volume	Plasma proteins (albumin) maintain osmotic pressure → prevent edema
Protection	Hemostasis	Platelets + clotting proteins → blood clot formation at injury site
Protection	Immunity	WBCs, antibodies, complement proteins fight pathogens

■ Why is blood a connective tissue?

All connective tissues have a matrix + suspended cells. Blood's matrix = plasma; cells = formed elements. Although fluid, it fits the structural definition perfectly.

§2 · Plasma

Plasma = ~55% of blood volume. Straw-colored, non-living liquid matrix.

Component	%	Examples / Notes
Water	90%	Primary solvent; maintains fluidity; absorbs heat
Plasma Proteins	~7%	Albumin (60%), globulins, fibrinogen — mainly liver-made
Electrolytes	~1%	Na ⁺ , K ⁺ , Ca ²⁺ , Mg ²⁺ , Cl ⁻ , HCO ₃ ⁻ ; pH & fluid homeostasis
Nutrients	<1%	Amino acids, glucose, lipids, vitamins (absorbed from GI tract)
Respiratory Gases	trace	O ₂ and CO ₂ dissolved in plasma
Hormones	trace	Thyroid, cortisol, EPO, insulin, etc.
Waste Products	trace	Urea, creatinine, BUN, uric acid → filtered by kidneys

Plasma Proteins — Detail

Protein	% of PP	Function	Source
Albumin	60%	Osmotic pressure; keeps water in vessels; prevents edema	Liver
Alpha/Beta Globulins	36%	Transport lipids, fat-soluble vitamins, metal ions	Liver
Gamma Globulins	part of 36%	Antibodies (IgG, IgM, etc.) — immune defense	Plasma cells (B-cells)
Fibrinogen	4%	Coagulation: fibrinogen → fibrin mesh (clot)	Liver
Clotting Factors	various	Factors VIII, IX, prothrombin → cascade	Liver (mostly)

■ Clinical: Kwashiorkor (Severe Protein Malnutrition)

↓ protein intake → ↓ albumin → ↓ osmotic pressure → water leaves vessels into interstitium → edema.

Classic 'pot belly' is fluid accumulation, NOT fat. Also: brittle hair, muscle wasting.

Treatment: gradual protein reintroduction. Never just give protein rapidly — refeeding syndrome risk.

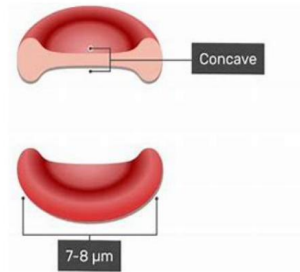
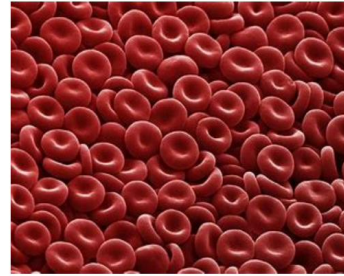
§3 · Red Blood Cells (Erythrocytes)

Structure

- **Biconcave disc** — 7–8 μm , flattened with depressed center
 - Maximizes surface area relative to volume → rapid gas exchange at every point
- **Anucleate** — no nucleus, no organelles
 - 97% of RBC volume = hemoglobin; no nucleus = can't divide or self-repair
- **Flexible membrane** — can twist, squeeze, deform
 - Passes through capillaries narrower than itself; rigid RBCs → microvascular problems
- **Anaerobic ATP production** — lacks mitochondria
 - Does NOT consume the O_2 it carries → 100% delivery efficiency

Structural Characteristics

- **Small biconcave disc** like cells
 - Flattened discs with depressed centers
 - Lighter in color in the center
- Mature erythrocytes lack a nucleus
 - **Anucleate with no organelles**
 - **Bags of hemoglobin**
- **Able to deform** as needed to go in and out of vessels
 - Twist, turn, become cup shaped
- Erythrocytes are the **major factor contributing to blood viscosity.**



▲ RBC biconcave disc structure and cross-section (7–8 μm)

Hemoglobin — Structure & Function

Hemoglobin (Hb) = 4 heme groups (each with central Fe^{2+} iron) + 4 globin protein chains.

Fact	Detail
O_2 per Hb molecule	4 O_2 molecules (one per heme/iron)
Hb per RBC	250 million hemoglobin molecules
O_2 per RBC	1 BILLION O_2 molecules per single RBC
Iron's role	Fe^{2+} binds O_2 reversibly; iron deficiency → IDA → fatigue
Why inside RBC?	Prevents Hb breakdown, leakage, and ↑ blood viscosity
Scarlet vs dark red	OxyHb = scarlet; DeoxyHb = dark red/maroon
Hb (Women/Men)	12–16 g/100 mL (W) · 13–18 g/100 mL (M)

Blood Viscosity

RBC count is the PRIMARY determinant of blood viscosity.

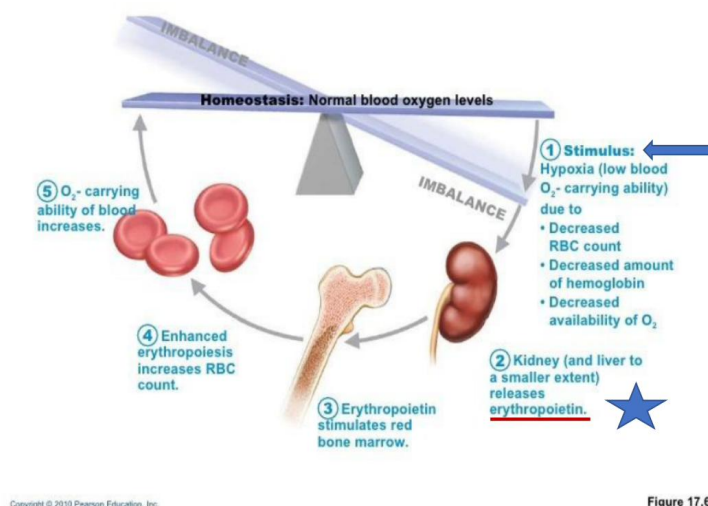
- ↑ RBCs → ↑ viscosity → slower flow → **risk of thrombus/clot**
- ↓ RBCs (anemia) → ↓ viscosity → faster flow → poor O_2 delivery

■ Blood Thinners (Anticoagulants)

Aspirin (81mg), heparin, warfarin — reduce platelet aggregation / clotting cascade.

They do NOT destroy RBCs. Prescribed for: DVT prevention, post-MI, Afib, high CV risk.

Erythropoiesis — RBC Production



▲ Erythropoietin feedback loop: hypoxia → kidneys → EPO → bone marrow → more RBCs

- Hemocytoblast:** Multipotential stem cell in RED bone marrow (axial skeleton + proximal humerus/femur)
- Myeloid pathway:** Commits to myeloid route (irreversible); cannot change to lymphoid
- Erythroblast:** Begins synthesizing hemoglobin; nucleus present and large
- Nucleus ejected:** Nucleus shrinks progressively → expelled when Hb level sufficient
- Reticulocyte:** Immature RBC; still has ribosomes; released into bloodstream
- Mature Erythrocyte:** Ribosomes degraded in ~2 days; fully formed biconcave disc; lives 100–120 days
- Destruction:** Aged RBCs → phagocytized by SPLEEN (primarily) + liver macrophages

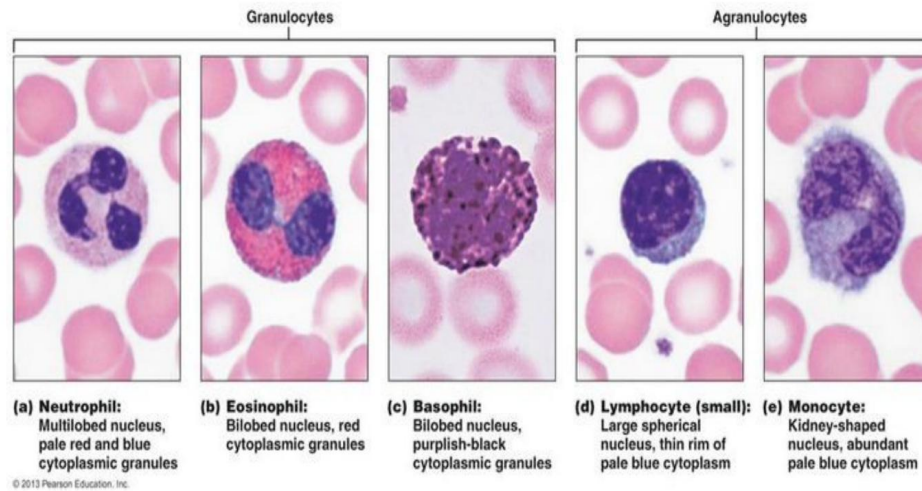
Key Stat	Value
Stem cell → reticulocyte	~15 days
Reticulocyte → mature RBC	~2 days (in bloodstream)
Normal reticulocyte %	1–2% of circulating RBCs
Elevated reticulocyte %	Bone marrow compensating (bleeding, hemolysis, post-iron therapy)
RBC lifespan	100–120 days
New blood per day	~100 billion cells / ~1 oz (28–30 mL)
Regulation hormone	Erythropoietin (EPO) — mainly kidneys, stimulated by hypoxia

Dietary Requirements for Erythropoiesis

Nutrient	Role	Deficiency → Anemia Type
Iron (Fe^{2+})	Center of heme; binds O_2	Iron Deficiency Anemia (IDA) — most common worldwide
Amino Acids	Build globin chains	Poor Hb synthesis; protein malnutrition
Vitamin B12	DNA synthesis for cell division	Pernicious anemia / megaloblastic anemia
Folate (B9)	DNA synthesis (esp. rapid division)	Megaloblastic anemia; neural tube defects (pregnancy)
Vitamin C	Enhances Fe absorption in GI	Reduced iron absorption → secondary anemia
EPO (hormone)	Signals bone marrow to produce RBCs	Renal anemia (kidneys can't make EPO)

§4 · White Blood Cells (Leukocytes)

Only TRUE cells in blood (nucleus + all organelles). <1% of blood volume but the body's mobile immune army. Bloodstream = transport highway to sites of infection.



▲ Granulocytes (N, E, B) vs Agranulocytes (Lymphocyte, Monocyte) — microscopy

Never Let Monkeys Eat Bananas

Neutrophils · Lymphocytes · Monocytes · Eosinophils · Basophils (high→low %)

WBC	% Blood	Category	Nucleus	Function
Neutrophils	50–70%	Granulocyte	Multi-lobed	Bacteria slayers; 1st responders; phagocytosis
Lymphocytes	25–33%	Agranulocyte	Large/round	T-cells: kill virus-infected cells; B-cells: antibodies
Monocytes	3–8%	Agranulocyte	Kidney-shaped	→ Macrophages in tissue; chronic infections/viruses
Eosinophils	2–4%	Granulocyte	Bi-lobed	Parasitic worms; allergies; asthma
Basophils	0–0.75%	Granulocyte	Bi-lobed	Histamine release (inflammation); contain heparin

WBC Mobilization

- **Chemotaxis:** WBCs follow chemical signals from damaged tissue to infection site
- **Diapedesis:** WBCs squeeze through vessel walls to reach tissue ('leaping across')
- **Leukocytosis:** ↑ WBC (infection/inflammation); can DOUBLE in 1 hour
- **Leukopenia:** ↓ WBC (chemotherapy, immunosuppression, bone marrow failure)

§5 · Platelets (Thrombocytes)

Feature	Detail
Origin	Cytoplasmic fragments of MEGAKARYOCYTES (largest bone marrow cell)
True cell?	No — anucleate cell fragments → cannot divide; age rapidly
Lifespan	8–10 days if unused; cleared by spleen
Count	150,000–500,000 per μL blood
Regulation hormone	Thrombopoietin (liver + kidneys)
Normal state	Kept INACTIVE by endothelial-secreted Nitric Oxide (NO)
Activated by	Exposed collagen fibers at vessel injury site
Contents	Granules with ADP, serotonin, thromboxane, clotting chemicals

§6 · Hemostasis

Hemostasis = stopping blood loss after vessel injury. Must be **fast, localized, controlled**.

Focus on ... Bone Marrow

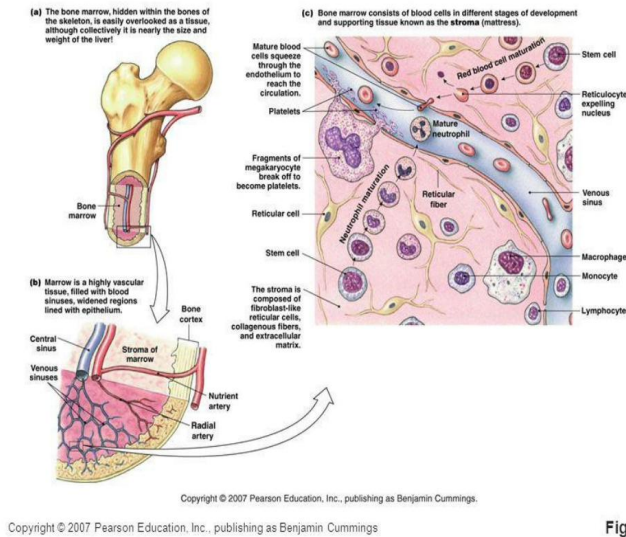


Figure 16-4a

▲ Bone marrow cross-section: RBC maturation, megakaryocyte → platelets, WBC formation

Step 1 · Vascular Spasm

Immediate vessel constriction → ↓ blood flow to injured area. Triggered by:

- Direct injury to vascular smooth muscle
- Chemicals from endothelial cells and platelets (serotonin, thromboxane)
- Pain receptor reflex signals

Step 2 · Platelet Plug Formation (~1 minute)

- Vessel damage exposes **collagen fibers** → platelets adhere strongly
- Platelets swell, form spike-like processes, become stickier
- Release **ADP** → recruits more platelets
- Release **serotonin + thromboxane** → reinforce vascular spasm
- Aggregate into loose platelet plug that slows/stops minor bleeding

Step 3 · Coagulation (Fibrin Mesh)

Cascade of plasma clotting factors (mostly liver-made) activates in sequence:

- **Prothrombin** (inactive) → **Thrombin** (active enzyme, via cascade)
- **Fibrinogen** (soluble) → **Fibrin** (insoluble threads, mesh)
- Fibrin mesh traps RBCs + platelets → stable, sealed blood clot
- Key cofactor: **Vitamin K** required for factors II, VII, IX, X synthesis

■ Acupuncture Clinical Pearl — Post-Needle Bleeding

After removing a needle, apply firm **STEADY PRESSURE** with cotton ball — do **NOT** massage.

Why: Platelet plug + fibrin mesh form within ~30–60 seconds but remain **FRAGILE**.

Massage disrupts the forming clot → re-bleeding → next-day bruising.

Rule: Press firmly 30 sec; if no bleeding, remove. Massage = broken clot = hematoma.

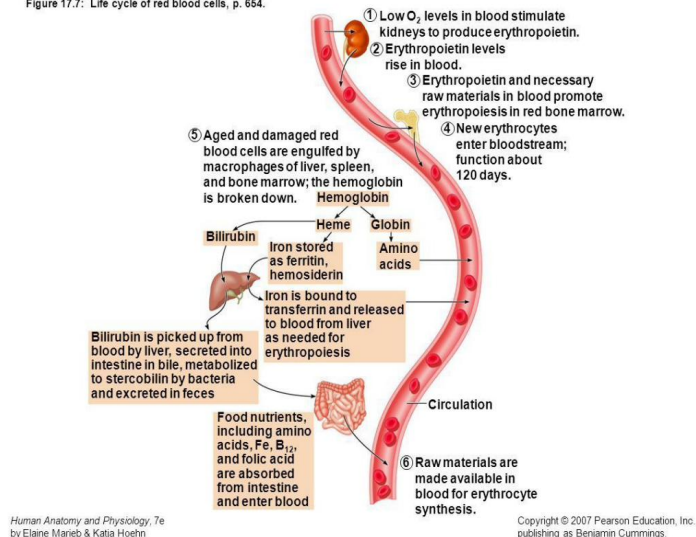
§7 · Disorders of Blood

Anemias — Reduced O₂-Carrying Capacity

Anemia = a **sign** of underlying pathology, not a disease itself. Symptoms: fatigue, pallor, dyspnea, palpitations, cold extremities.

Type	Mechanism	Notes
Iron Deficiency (IDA)	↓ Fe → ↓ heme → ↓ Hb → microcytic hypochromic RBCs	Most common; brittle nails, spoon nails, pica
Pernicious Anemia	Lack of Intrinsic Factor → ↓ B12 absorption	Large fragile RBCs; neurological symptoms (B12 needed for myelin)
Aplastic Anemia	Bone marrow failure → no stem cell production	Pancytopenia (all cell lines low); very serious; idiopathic or radiation
Sickle Cell Anemia	Genetic HbS mutation → sickle-shaped rigid RBCs	Vaso-occlusive crisis; hemolysis; autosomal recessive
Thalassemia	Genetic defect in globin chain synthesis	Mediterranean, African, SE Asian; target cells on smear
Hemolytic Anemia	Premature RBC destruction	Jaundice, splenomegaly; autoimmune or chemical triggers
Renal Anemia	Kidneys fail → ↓ EPO → ↓ RBC production	Normocytic; treated with EPO injections
Blood Loss Anemia	Trauma, cancer, menstruation	Acute vs chronic; compensatory reticulocytosis

Figure 17.7: Life cycle of red blood cells, p. 654.



▲ RBC life cycle: EPO regulation, nutrient absorption, Hb breakdown pathway

Polycythemia — Excess RBCs

- **Polycythemia Vera**: bone marrow malignancy → uncontrolled RBC overproduction
- **Secondary**: physiological response to high altitude (↓ O₂ → ↑ EPO)
- **Blood doping**: athletes inject exogenous EPO (banned)
- Risk: ↑ viscosity → sluggish flow → ↑ thrombus, stroke, MI risk

Hemostatic Disorders

Disorder	Type	Mechanism	Clinical
DVT	Thrombus	Immobility → blood pools → platelet aggregate in intact vessel	Long flights, bedrest; walk to prevent; treat with heparin/warfarin
Pulmonary Embolism	Embolus	DVT dislodges → travels to pulmonary vessels → blocks blood flow	Life-threatening; sudden dyspnea, chest pain, hypoxia

Disorder	Type	Mechanism	Clinical
Stroke (CVA)	Embolus	Embolus blocks cerebral vessel	FAST: Face/Arm/Speech/Time
Thrombocytopenia	Bleeding	↓ platelet count (autoimmune, drug, marrow failure)	Petechiae, easy bruising, prolonged bleeding time
Hemophilia A/B	Bleeding	Factor VIII (A) or IX (B) deficiency → cascade fails	X-linked recessive; treat with factor replacement
Liver Disease	Bleeding	Liver fails to make clotting factors (fibrinogen, prothrombin)	Cirrhosis, end-stage liver disease; ↑ PT/INR

Anticoagulant Drugs

Drug	Mechanism	Route / Clinical Use
Aspirin (81–325 mg)	Inhibits COX → blocks thromboxane A ₂ → ↓ platelet aggregation	Oral; daily prophylaxis in CV risk patients
Heparin	Enhances Antithrombin III → inhibits thrombin + Factor Xa	IV/SQ; hospital; acute DVT, PE, post-surgical
Warfarin (Coumadin)	Blocks Vitamin K → ↓ synthesis of factors II, VII, IX, X	Oral; long-term anticoagulation after discharge
Note: Vitamin K	Required for production of clotting factors II, VII, IX, X	Newborns get Vit K injection at birth (no gut bacteria yet)

§8 · TCM & Clinical Connections

Blood (Xue 血) in TCM

Blood is a dense, nourishing Yin substance that moistens tissues, nourishes the Shen (mind), and circulates through vessels guided by Heart Qi. Xue Xu (Blood deficiency) maps closely to Western anemia: pallor, fatigue, poor memory, dizziness, palpitations, dry skin.

Spleen Qi & Iron Metabolism

The Spleen (Pi 脾) transforms food essence (Gu Qi) into Blood. Iron deficiency anemia often correlates with Spleen Qi Deficiency: fatigue, poor digestion, loose stools, pale complexion. Treatment: tonify Spleen (ST-36, SP-6, SP-3, BL-20) + dietary iron + Vit C.

Kidney Essence (Jing) & Bone Marrow

TCM states 'Kidney governs bones and produces marrow.' This directly parallels Western hematopoiesis — bone marrow makes blood cells. Renal anemia (↓ EPO from diseased kidneys) corresponds to Kidney deficiency. Points: KD-3, BL-23, GV-4.

Blood Stasis (Yu Xue 瘀血) & Thrombosis

Excess clotting or sluggish circulation → Blood Stasis in TCM. DVT, polycythemia, and atherosclerosis all have Blood Stasis components. Points: SP-10 (Xuehai 血海 = Sea of Blood), BL-17, LV-3, PC-6. Herbs: Dan Shen (Salvia), Hong Hua, Tao Ren.

DVT Prevention — Qi & Blood Movement

Long immobility → Qi and Blood stagnation in the lower extremities (channel blockage). TCM: movement, elevation, moxa, and Blood-invigorating herbs. Western: walk during long flights; ankle circles; compression stockings.

Hemostasis & Acupuncture Practice

Understanding the 3-step hemostasis process directly informs needle technique: vessel spasm (Step 1) begins immediately; press-not-massage to allow Step 2 platelet plug and Step 3 fibrin mesh to complete undisturbed. Clinical skill, not just textbook knowledge.

§9 · Quiz Bank — Exam Prep

Key Mnemonics

NEVER LET MONKEYS EAT BANANAS

WBC high→low: Neutrophils · Lymphocytes · Monocytes · Eosinophils · Basophils

3 – 3 – 2 FUNCTIONS

3 Distributing (O₂, waste, hormones) · 3 Regulating (temp, pH, volume) · 2 Protecting

V – P – C HEMOSTASIS

Vascular spasm → Platelet plug → Coagulation (fibrin mesh)

BLAST = YOUNG · CYTE = MATURE · POIESIS = PRODUCTION

Hemocytoblast (young stem) → Erythrocyte (mature RBC) · Hematopoiesis = blood production

1 Hb = 4 O₂ · 1 RBC = 250M Hb · ∴ 1 RBC = 1 BILLION O₂

We are all billionaires — in oxygen. (Dr. Sadrolsadot)

◆ Check Your Understanding

1. List the 8 functions of blood across 3 categories.
2. Why is blood classified as connective tissue? What are its matrix and cellular components?
3. Are plasma proteins used as cellular fuel? Explain why/why not.
4. What are the 4 major plasma proteins and their functions?
5. Explain how albumin deficiency leads to edema. Name a clinical example.
6. What is the pH of blood and what maintains it?

Blood Overview & Plasma

◆ Check Your Understanding

1. Describe the structural features of an RBC and explain how each serves its function.
2. Calculate: How many O₂ molecules can one RBC carry? (Show math.)
3. Why do RBCs generate ATP anaerobically? What is the advantage?
4. Trace erythropoiesis step by step from stem cell to mature RBC.
5. What hormone regulates RBC production? Where is it made? What stimulates it?
6. What does a reticulocyte count of 10% indicate clinically?
7. List all dietary requirements for erythropoiesis and what happens when each is deficient.

Erythrocytes

◆ Check Your Understanding

1. What is the difference between granulocytes and agranulocytes? Name all 5 WBC types.
2. Which WBC becomes a macrophage in tissue? What does it fight?
3. Which WBC responds to parasitic infection? To allergies?
4. Which WBC releases histamine? What is histamine's role?
5. Explain diapedesis and chemotaxis.
6. What is leukocytosis? Leukopenia? What causes each?
7. How are platelets formed? From what precursor cell? What hormone regulates this?
8. What keeps platelets inactive normally? What activates them?

WBCs & Platelets

◆ Check Your Understanding

1. Describe the 3 steps of hemostasis in sequence.

2. What is the difference between fibrinogen and fibrin? Between prothrombin and thrombin?
3. Define thrombus vs embolus. When does an embolus become dangerous?
4. What is DVT? What causes it? What are the risks if the clot dislodges?
5. Explain the mechanism of: (a) aspirin, (b) heparin, (c) warfarin.
6. Why is Vitamin K injected into newborns at birth?
7. Why do we press (not massage) after removing an acupuncture needle?
8. Name 3 causes of anemia with distinct mechanisms.
9. What is polycythemia vera? Why is it dangerous?
10. What causes hemophilia A? B? How is it treated?

Hemostasis & Disorders

■■■■ · 'Blood flows without ceasing' · Huang Di Nei Jing